KnotBench-Photo real_v1: capture kit

1. Tie each cable from its sheet (next pages). Follow the drawing: a gap means
   the strand passes UNDER. Join the two ends (barrel connector, heat-shrink or tape)
   so the cable is a closed loop; for links tie two cables as drawn.
2. Before the first photo of a cable, take one reference shot of the cable next
   to its sheet (name it cXX_reference.jpg; it is excluded from evaluation).
3. For each cable, take 3 views in each of its 2 sessions and name the files exactly
   as in shot_list.csv (cXX_sY_vZ.jpg): 20 cables x 2 sessions x 3 views = 120 photos.
   Re-lay the cable loosely between sessions; do not re-tie it.
4. Annotate in tools/photo_annotator.html: load cables.csv once, then the photos;
   the file name fills cable, session, view and type. Click the cable width,
   then every crossing (centre, a point on the top strand, a point on the bottom
   strand) and set visibility.
5. Put the photos in data/knotbench_photo/real_v1/images/ and records.jsonl next
   to them. Run scripts/check_real_photos.py, then score ONCE:
   python scripts/eval_real_photos.py data/knotbench_photo/real_v1

Sessions (each is a different place, light and camera):

   sA: Desk by a window: daylight, matte light background (paper or wall), phone 1

   sB: Wooden floor or table: one desk lamp from the side (hard shadows), phone 2

   sC: Dark cloth or mat: dim room light, camera or phone 1 without flash

   sD: Cluttered desk (cables, papers, objects nearby): mixed light, phone 2

Views:

   v1: top-down, cable fills about two thirds of the frame

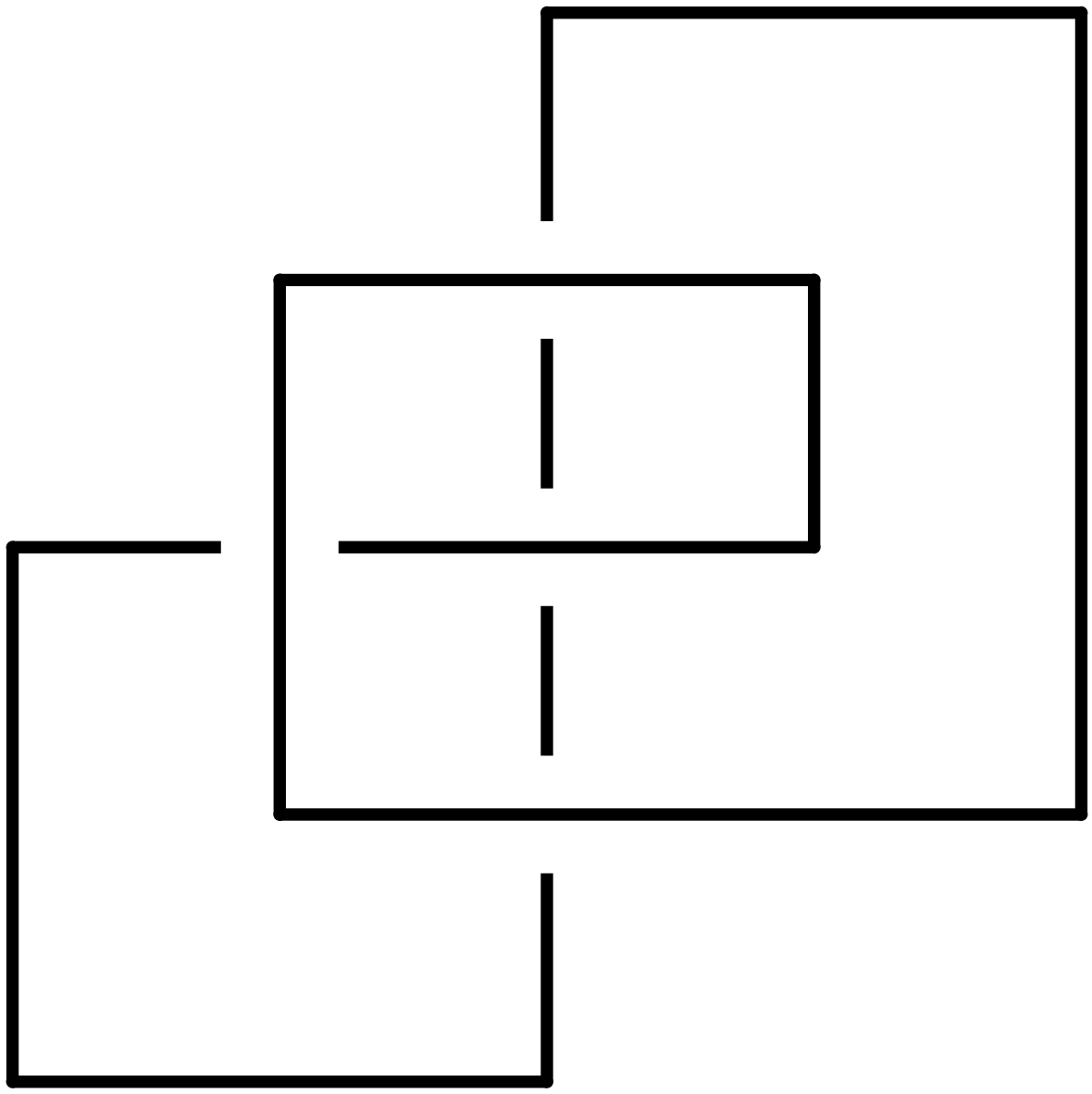
   v2: about 30 degrees tilt, from one side

   v3: about 60 degrees tilt, from the opposite side

Do not show these photos to anyone changing the model: the model is frozen
(results/frozen_model.json) and the scoring script logs every run.

c01: unknot (tangled, no knot)

type 0_1 | material: coax RG6, black, about 6.9 mm | sessions: sA, sB

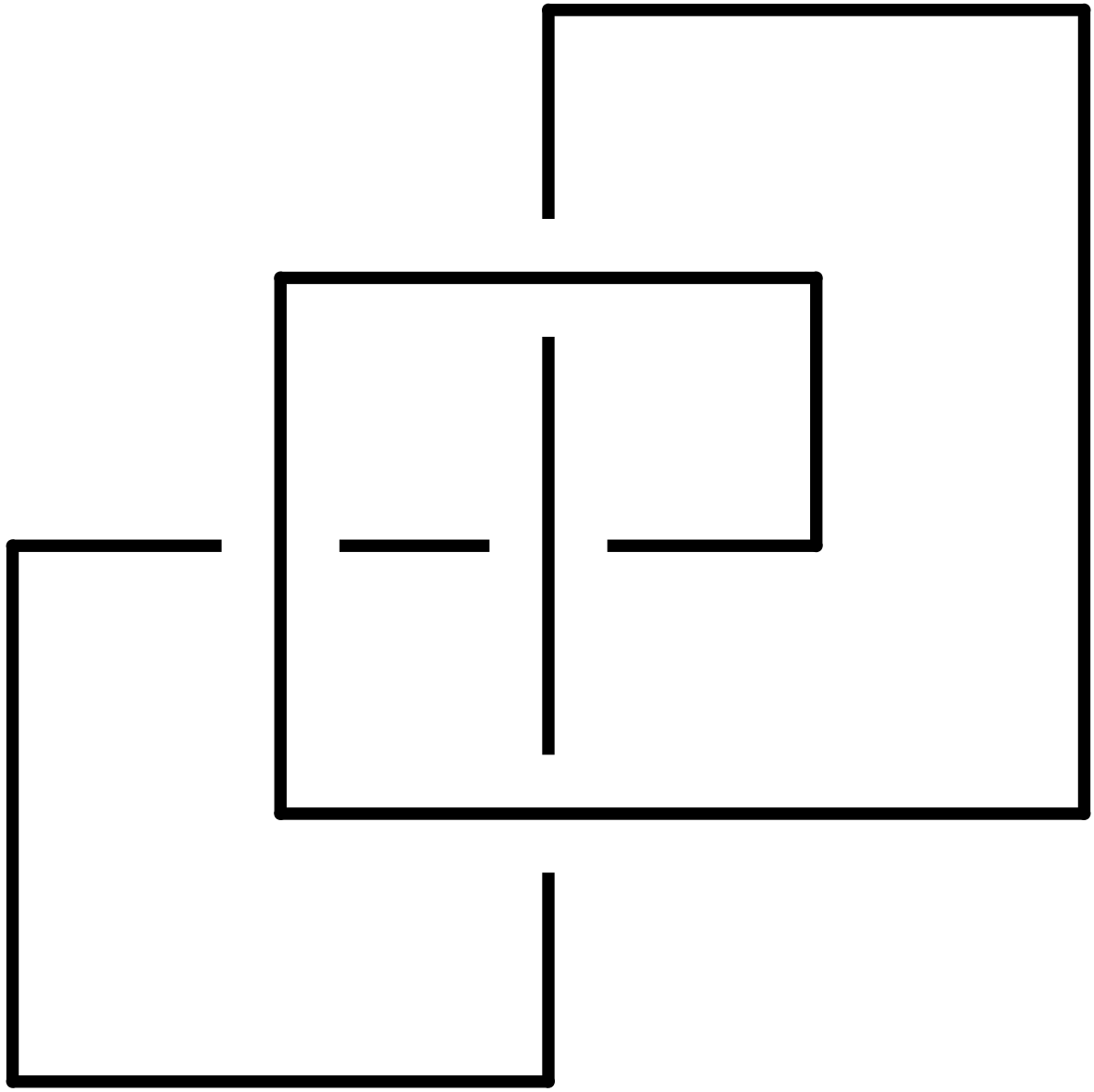


Gap = this strand passes UNDER the other one. Join the ends so the loop is closed.

Photos: c01_sA_v1.jpg, c01_sA_v2.jpg, c01_sA_v3.jpg, c01_sB_v1.jpg, c01_sB_v2.jpg, c01_sB_v3.jpg

c02: unknot (tangled, no knot)

type 0_1 | material: patch cable (Cat6), blue, about 6.0 mm | sessions: sC, sD

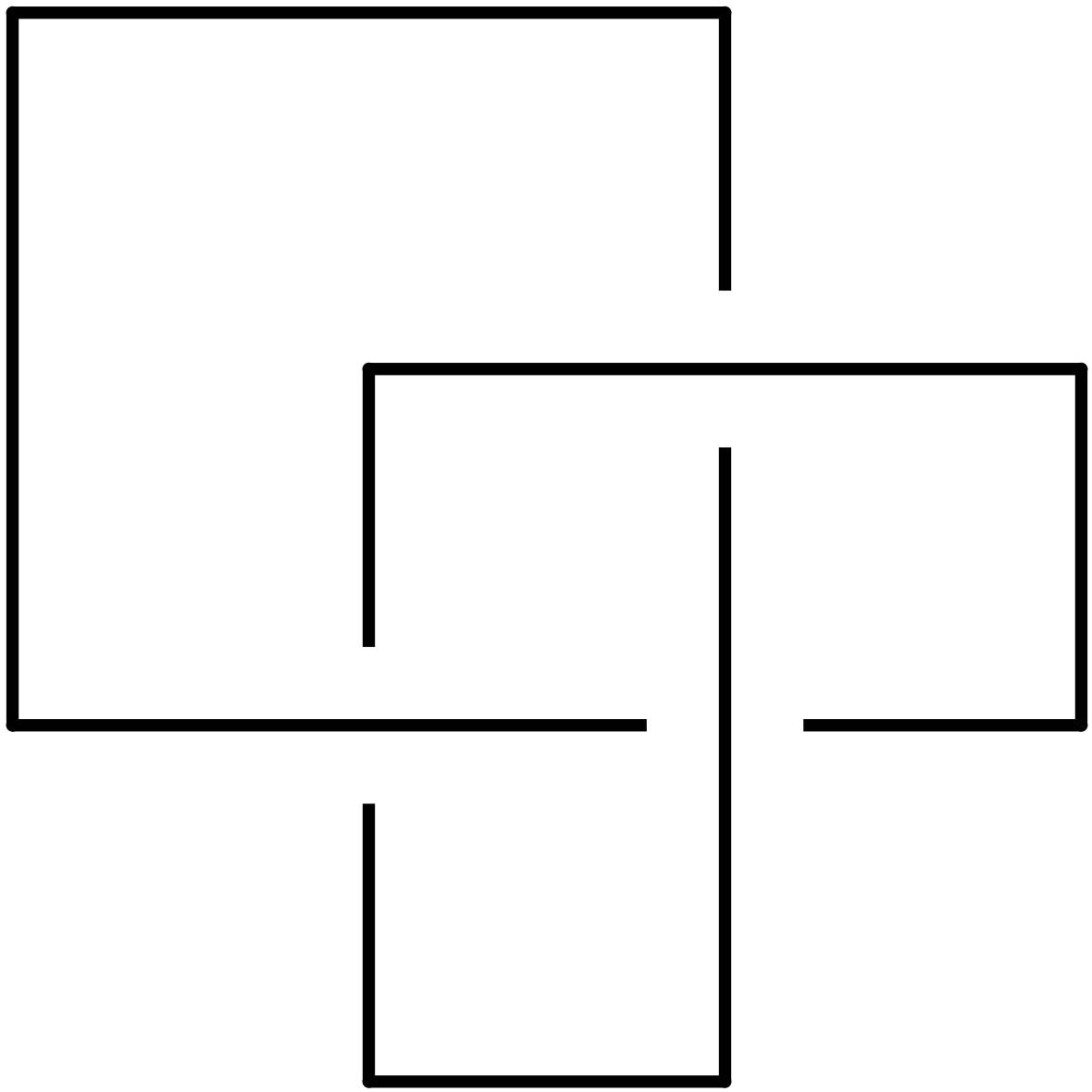


Gap = this strand passes UNDER the other one. Join the ends so the loop is closed.

Photos: c02_sC_v1.jpg, c02_sC_v2.jpg, c02_sC_v3.jpg, c02_sD_v1.jpg, c02_sD_v2.jpg, c02_sD_v3.jpg

c03: trefoil

type 3_1* | material: coax RG6, black, about 6.9 mm | sessions: sA, sC

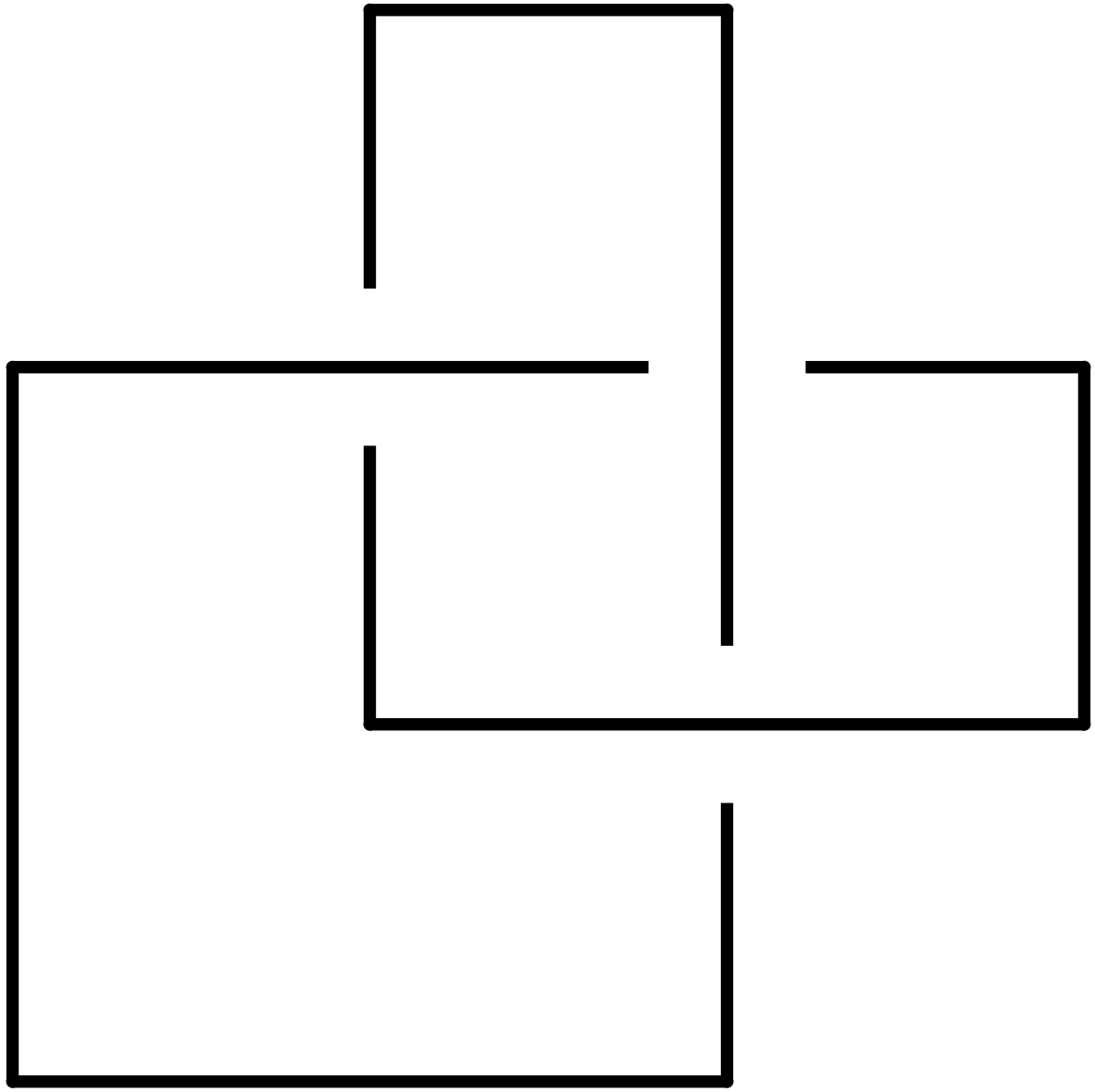


Gap = this strand passes UNDER the other one. Join the ends so the loop is closed.

Photos: c03_sA_v1.jpg, c03_sA_v2.jpg, c03_sA_v3.jpg, c03_sC_v1.jpg, c03_sC_v2.jpg, c03_sC_v3.jpg

c04: trefoil, mirror image

type 3_1 | material: rope 8 mm, any, about 8.0 mm | sessions: sB, sD

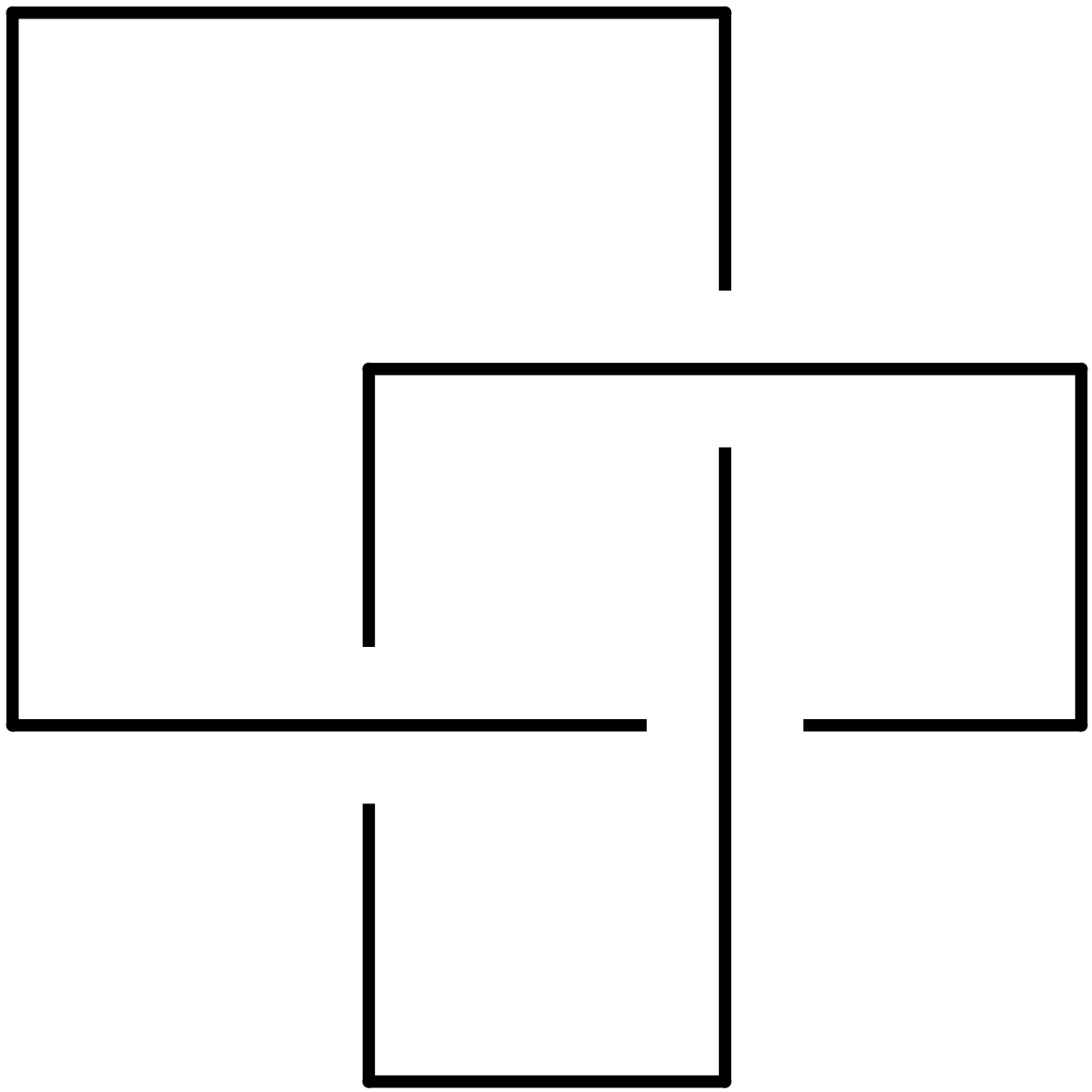


Gap = this strand passes UNDER the other one. Join the ends so the loop is closed.

Photos: c04_sB_v1.jpg, c04_sB_v2.jpg, c04_sB_v3.jpg, c04_sD_v1.jpg, c04_sD_v2.jpg, c04_sD_v3.jpg

c05: trefoil

type 3_1* | material: fishing line or thick thread, any, about 1.0 mm | sessions: sA, sD

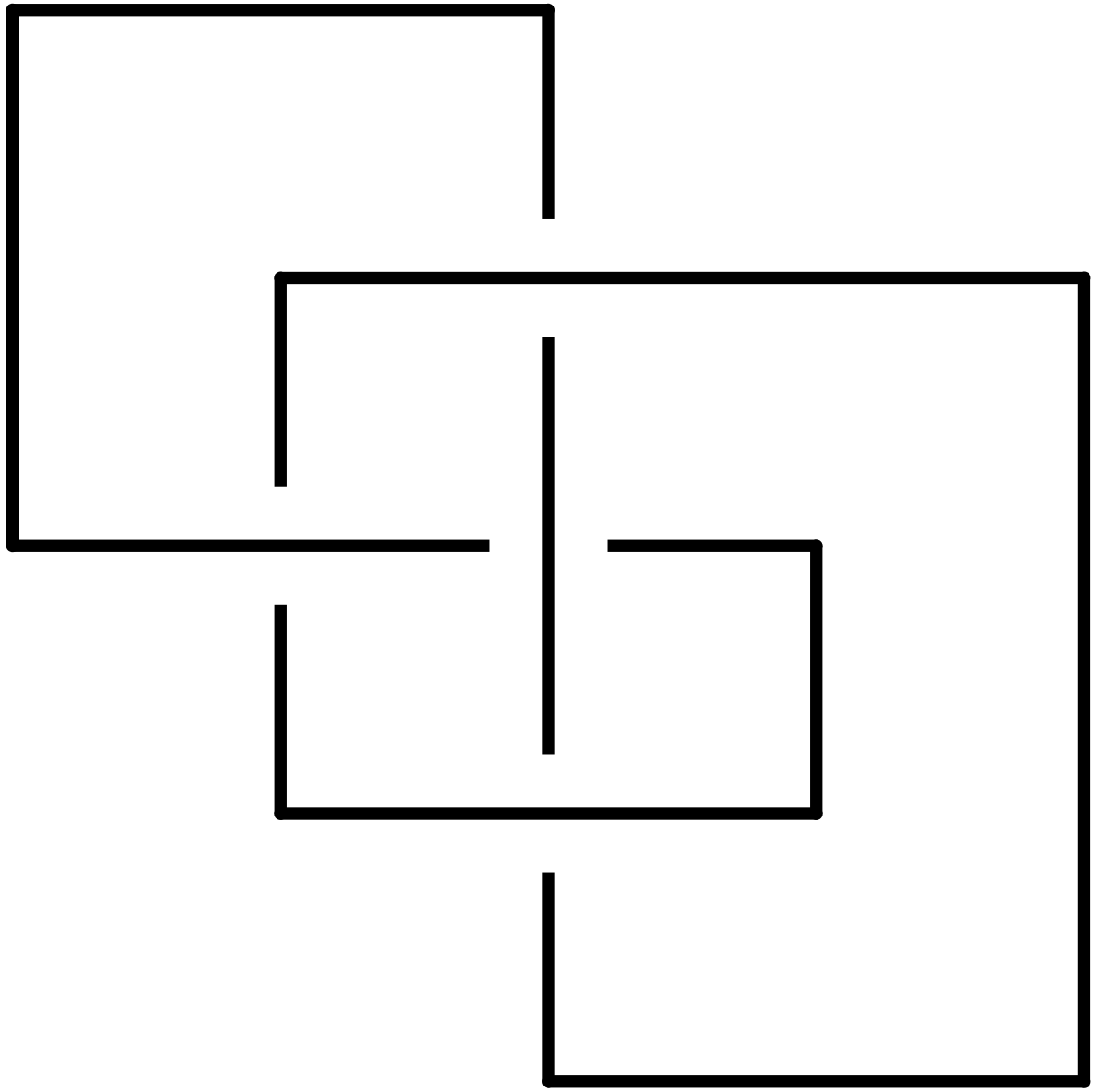


Gap = this strand passes UNDER the other one. Join the ends so the loop is closed.

Photos: c05_sA_v1.jpg, c05_sA_v2.jpg, c05_sA_v3.jpg, c05_sD_v1.jpg, c05_sD_v2.jpg, c05_sD_v3.jpg

c06: figure-eight knot

type 4_1 | material: coax RG58, black, about 5.0 mm | sessions: sB, sC

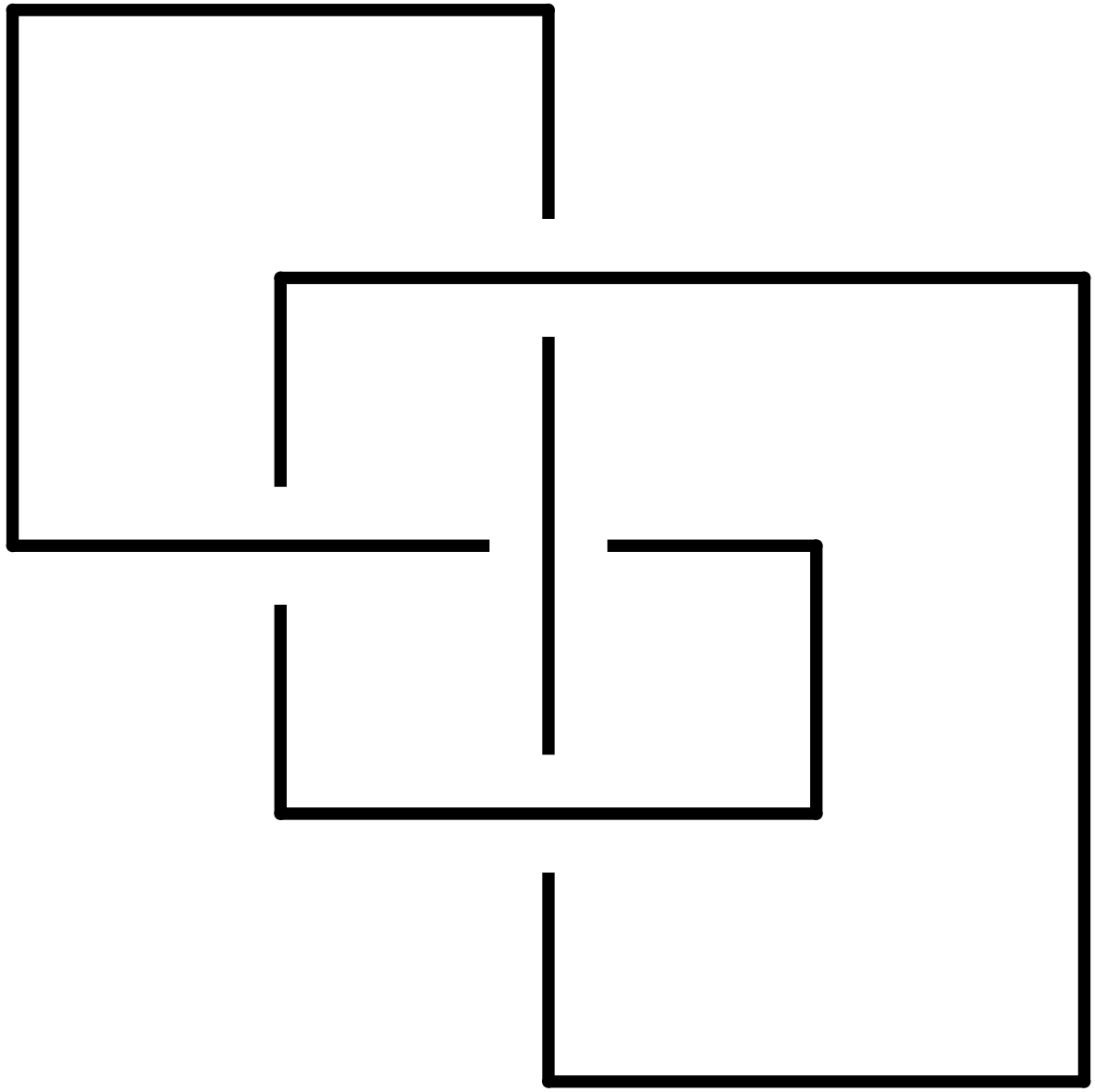


Gap = this strand passes UNDER the other one. Join the ends so the loop is closed.

Photos: c06_sB_v1.jpg, c06_sB_v2.jpg, c06_sB_v3.jpg, c06_sC_v1.jpg, c06_sC_v2.jpg, c06_sC_v3.jpg

c07: figure-eight knot

type 4_1 | material: rope 8 mm, any, about 8.0 mm | sessions: sA, sB

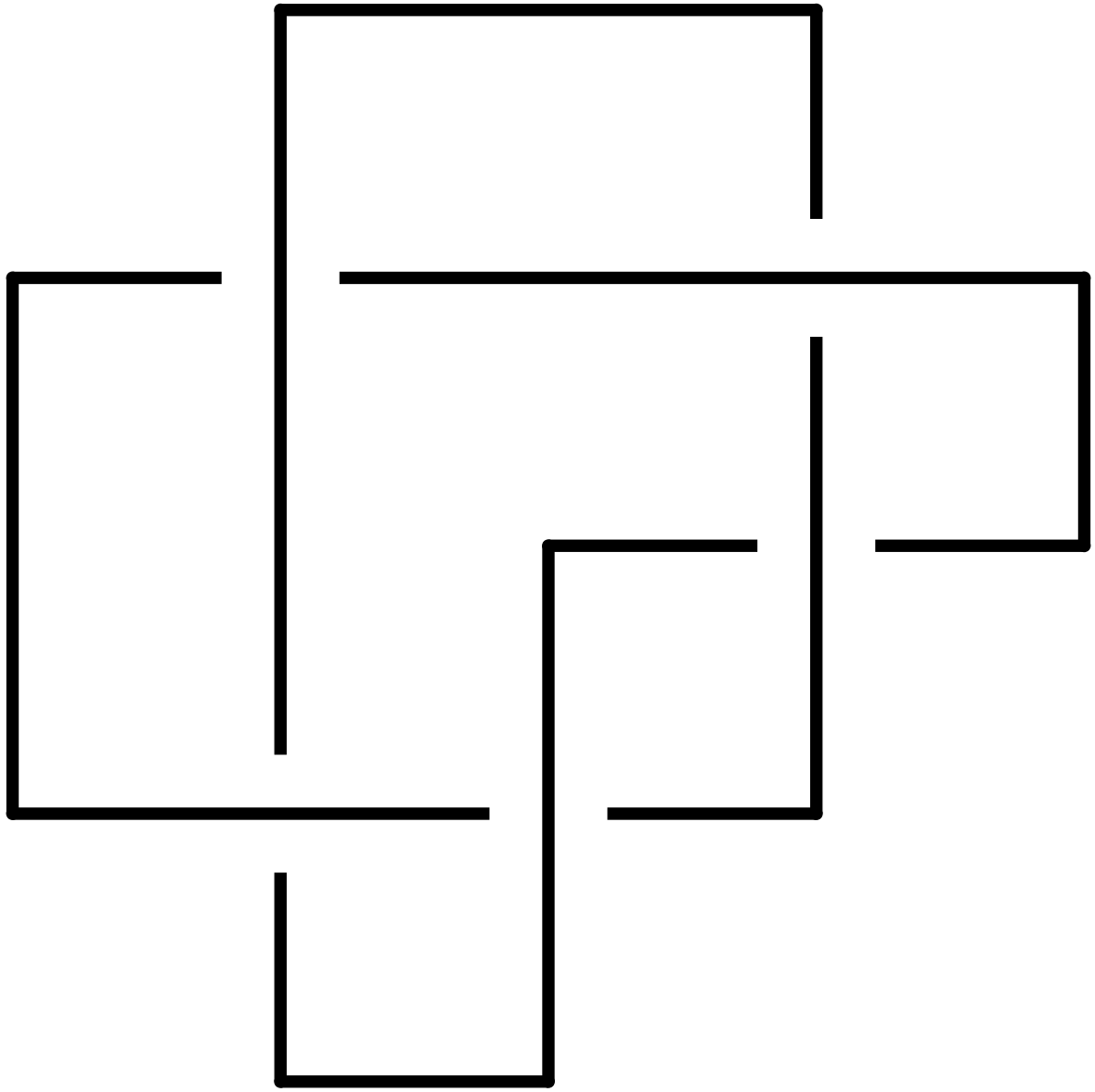


Gap = this strand passes UNDER the other one. Join the ends so the loop is closed.

Photos: c07_sA_v1.jpg, c07_sA_v2.jpg, c07_sA_v3.jpg, c07_sB_v1.jpg, c07_sB_v2.jpg, c07_sB_v3.jpg

c08: cinquefoil (5_1)

type 5_1* | material: patch cable (Cat6), any, about 6.0 mm | sessions: sC, sD

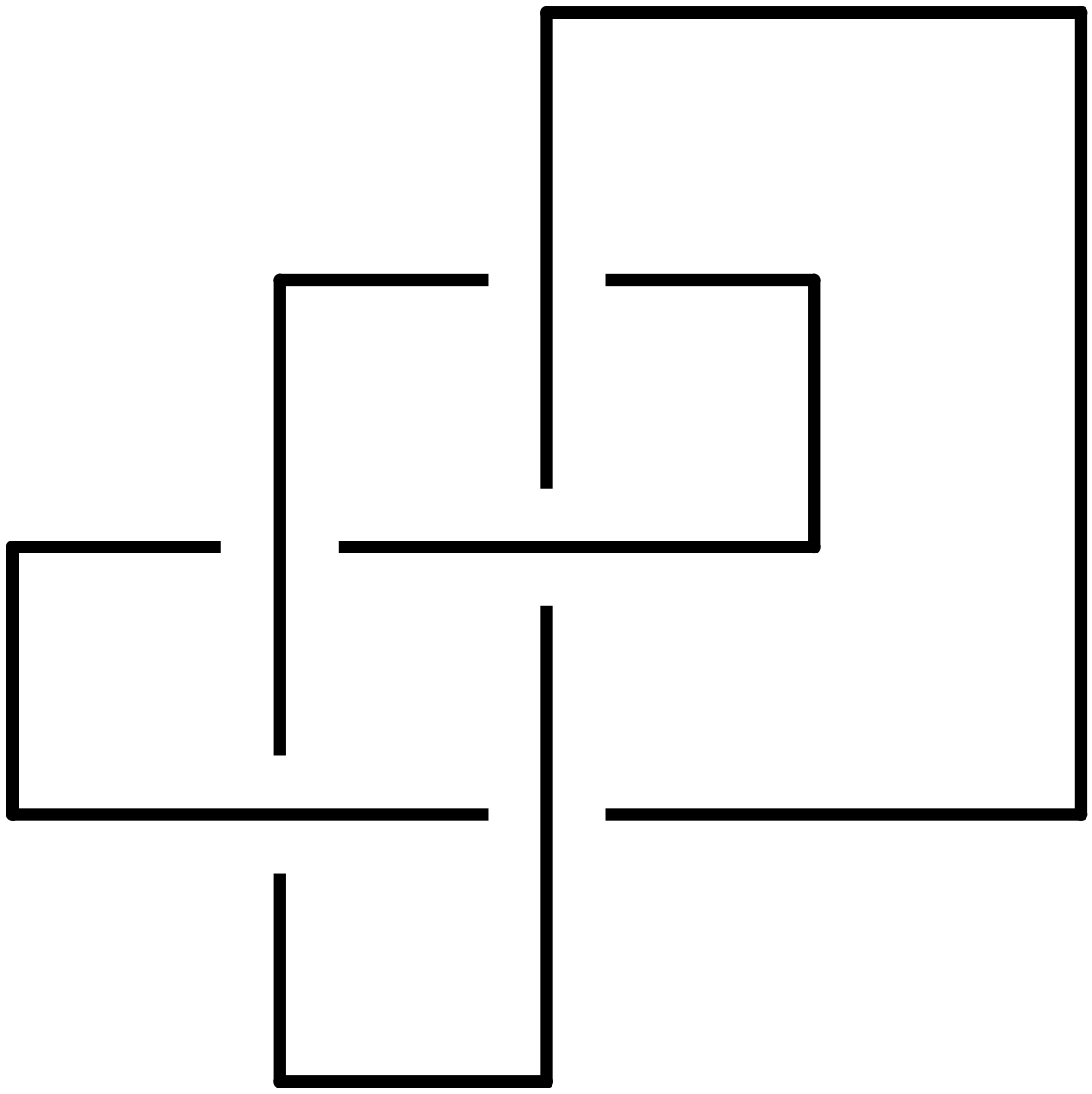


Gap = this strand passes UNDER the other one. Join the ends so the loop is closed.

Photos: c08_sC_v1.jpg, c08_sC_v2.jpg, c08_sC_v3.jpg, c08_sD_v1.jpg, c08_sD_v2.jpg, c08_sD_v3.jpg

c09: three-twist knot (5_2)

type 5_2 | material: coax RG6, white, about 6.9 mm | sessions: sA, sC

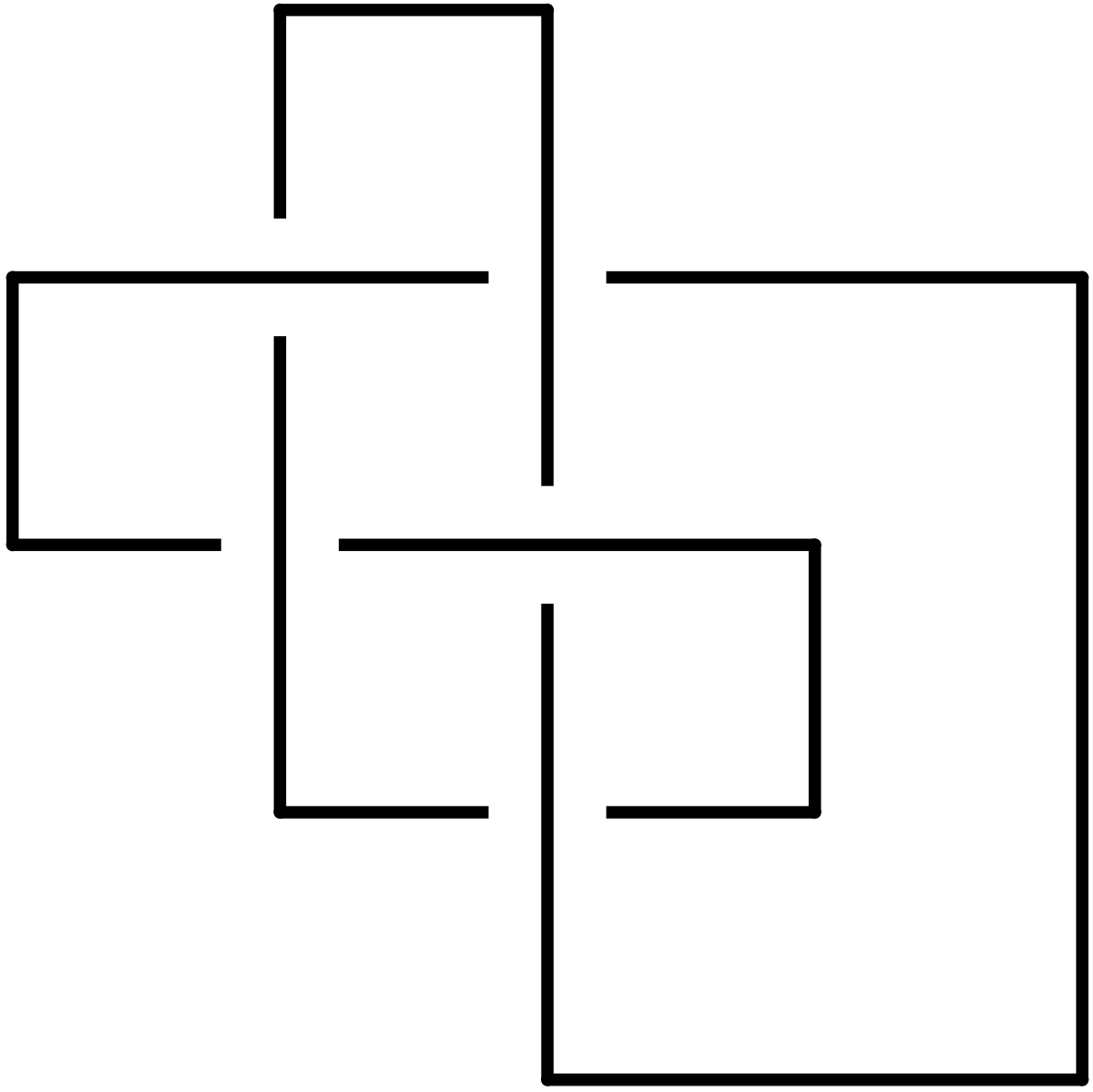


Gap = this strand passes UNDER the other one. Join the ends so the loop is closed.

Photos: c09_sA_v1.jpg, c09_sA_v2.jpg, c09_sA_v3.jpg, c09_sC_v1.jpg, c09_sC_v2.jpg, c09_sC_v3.jpg

c10: three-twist knot, mirror image

type 5_2* | material: rope 8 mm, any, about 8.0 mm | sessions: sB, sD

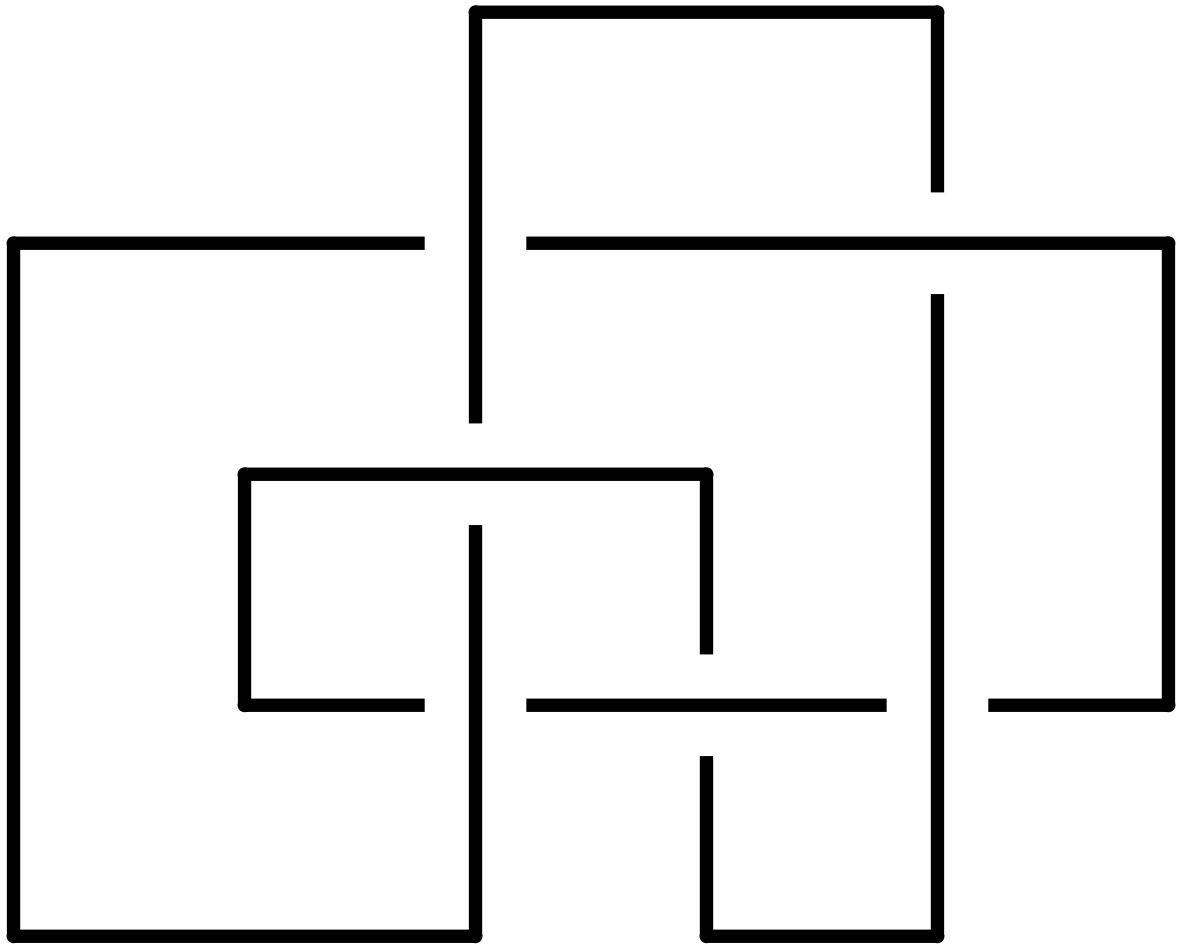


Gap = this strand passes UNDER the other one. Join the ends so the loop is closed.

Photos: c10_sB_v1.jpg, c10_sB_v2.jpg, c10_sB_v3.jpg, c10_sD_v1.jpg, c10_sD_v2.jpg, c10_sD_v3.jpg

c11: stevedore knot (6_1)

type 6_1 | material: coax RG58, black, about 5.0 mm | sessions: sA, sD

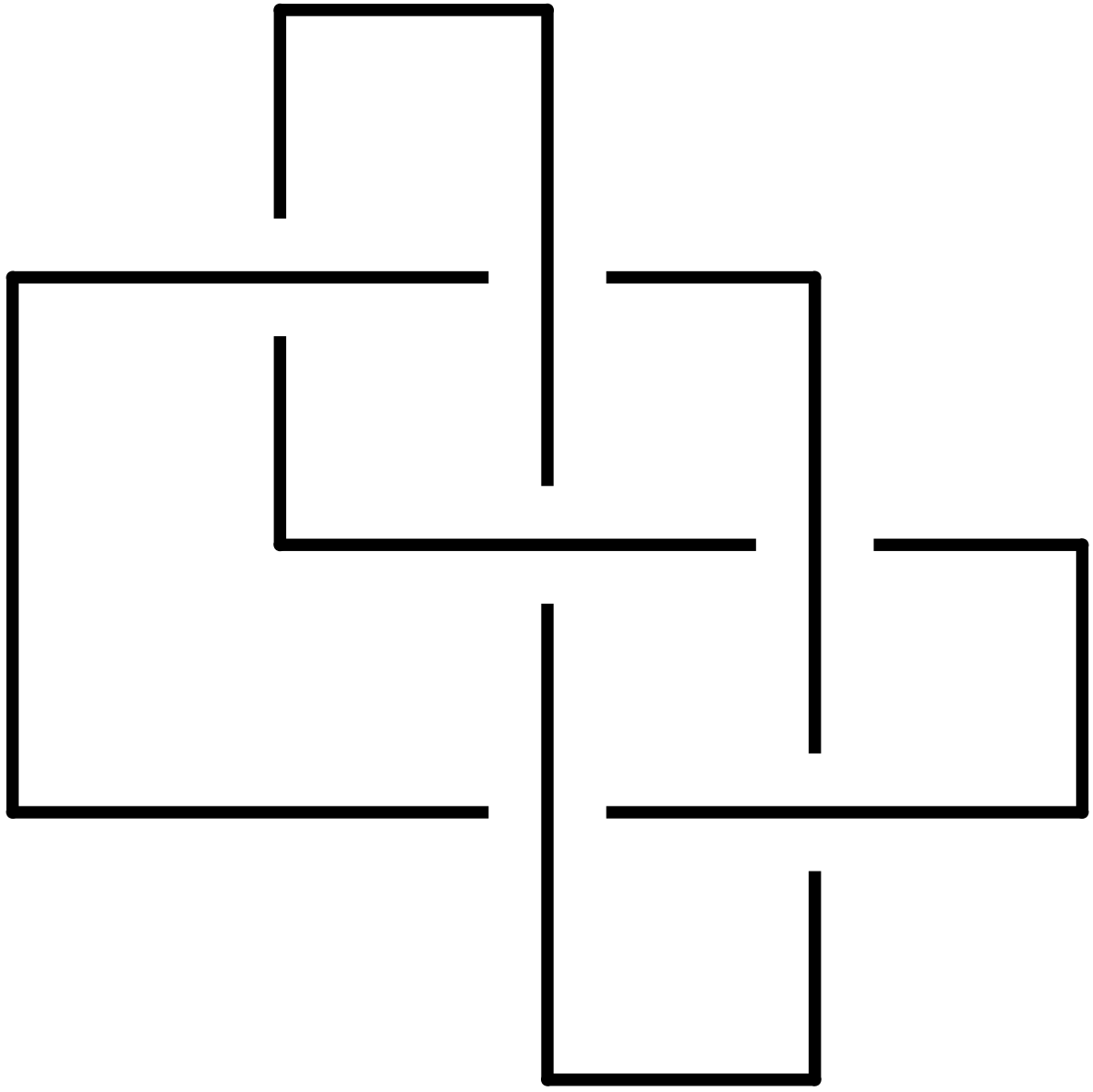


Gap = this strand passes UNDER the other one. Join the ends so the loop is closed.

Photos: c11_sA_v1.jpg, c11_sA_v2.jpg, c11_sA_v3.jpg, c11_sD_v1.jpg, c11_sD_v2.jpg, c11_sD_v3.jpg

c12: 6_2

type 6_2 | material: patch cable (Cat6), any, about 6.0 mm | sessions: sB, sC

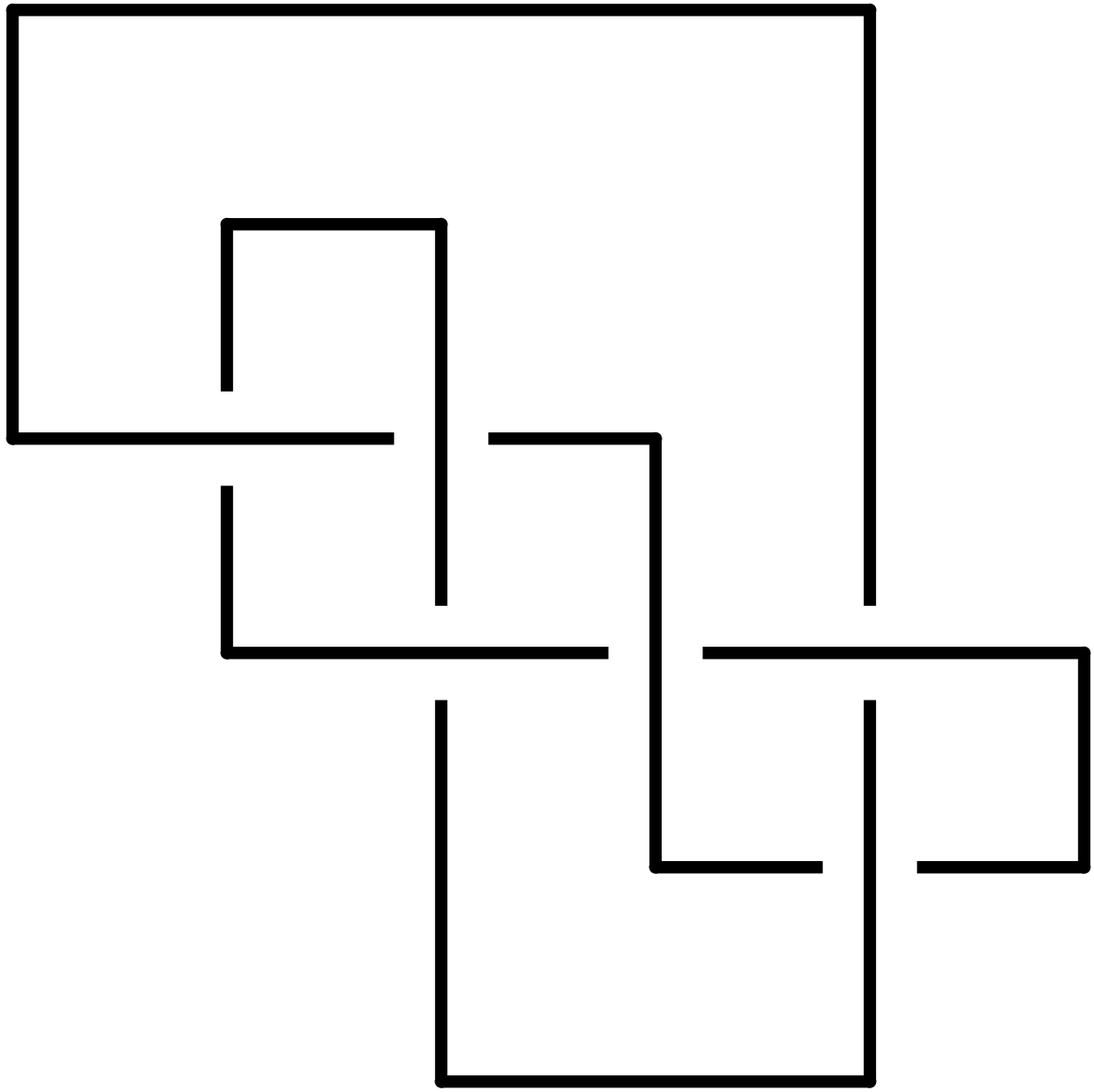


Gap = this strand passes UNDER the other one. Join the ends so the loop is closed.

Photos: c12_sB_v1.jpg, c12_sB_v2.jpg, c12_sB_v3.jpg, c12_sC_v1.jpg, c12_sC_v2.jpg, c12_sC_v3.jpg

c13: 6_3

type 6_3 | material: rope 8 mm, any, about 8.0 mm | sessions: sA, sB

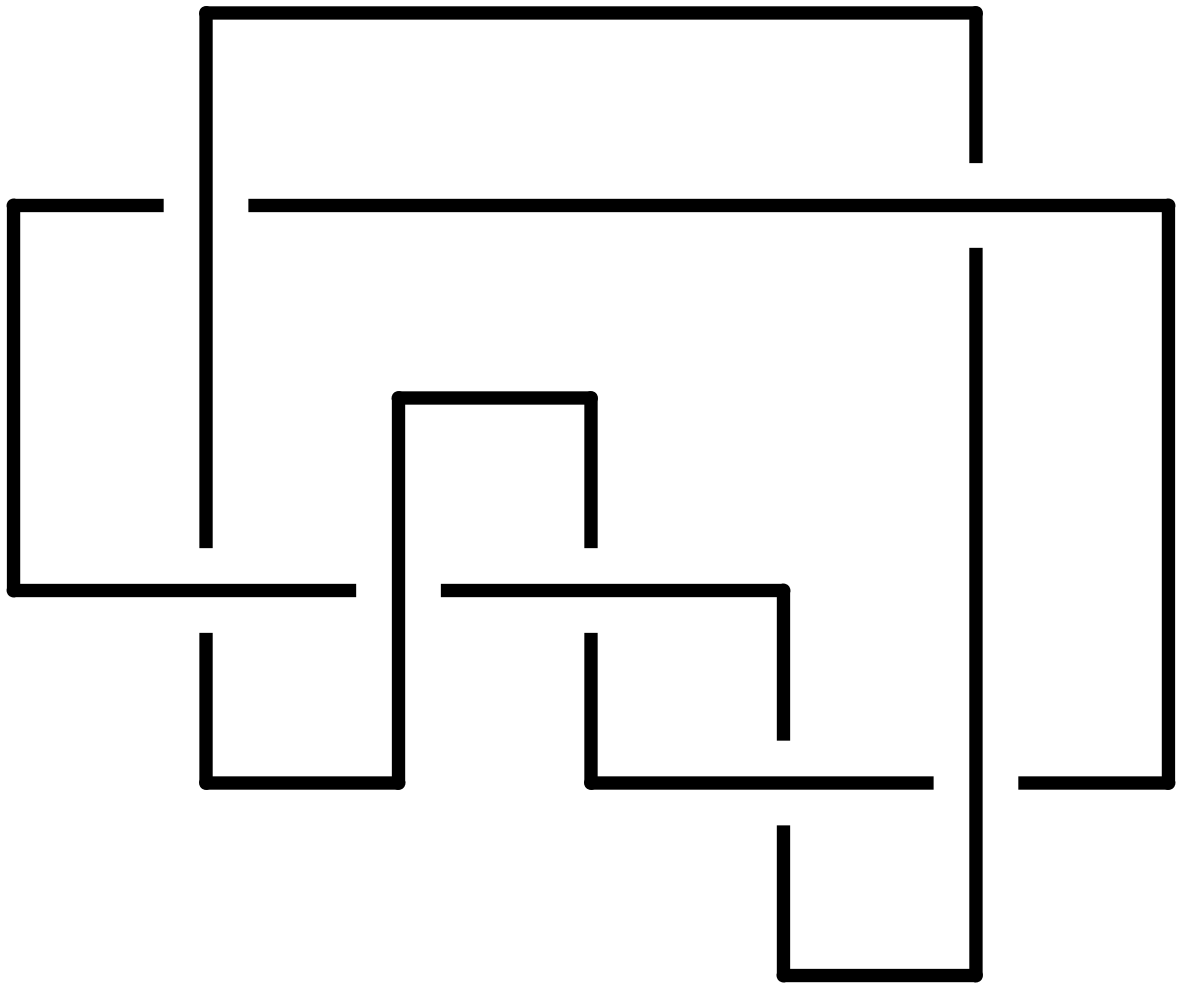


Gap = this strand passes UNDER the other one. Join the ends so the loop is closed.

Photos: c13_sA_v1.jpg, c13_sA_v2.jpg, c13_sA_v3.jpg, c13_sB_v1.jpg, c13_sB_v2.jpg, c13_sB_v3.jpg

c14: 7_1 (seven-crossing torus knot)

type 7_1* | material: fishing line or thick thread, any, about 1.0 mm | sessions: sC, sD

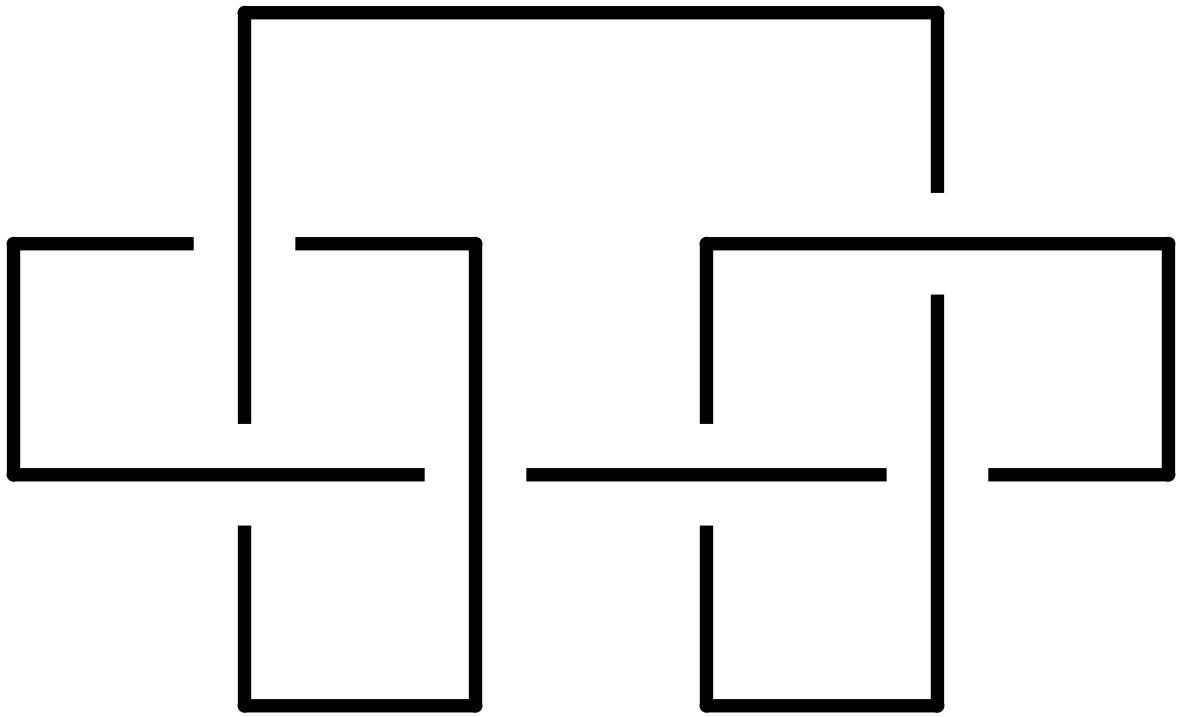


Gap = this strand passes UNDER the other one. Join the ends so the loop is closed.

Photos: c14_sC_v1.jpg, c14_sC_v2.jpg, c14_sC_v3.jpg, c14_sD_v1.jpg, c14_sD_v2.jpg, c14_sD_v3.jpg

c15: granny knot

type 3_1 # 3_1 | material: rope 8 mm, any, about 8.0 mm | sessions: sA, sC

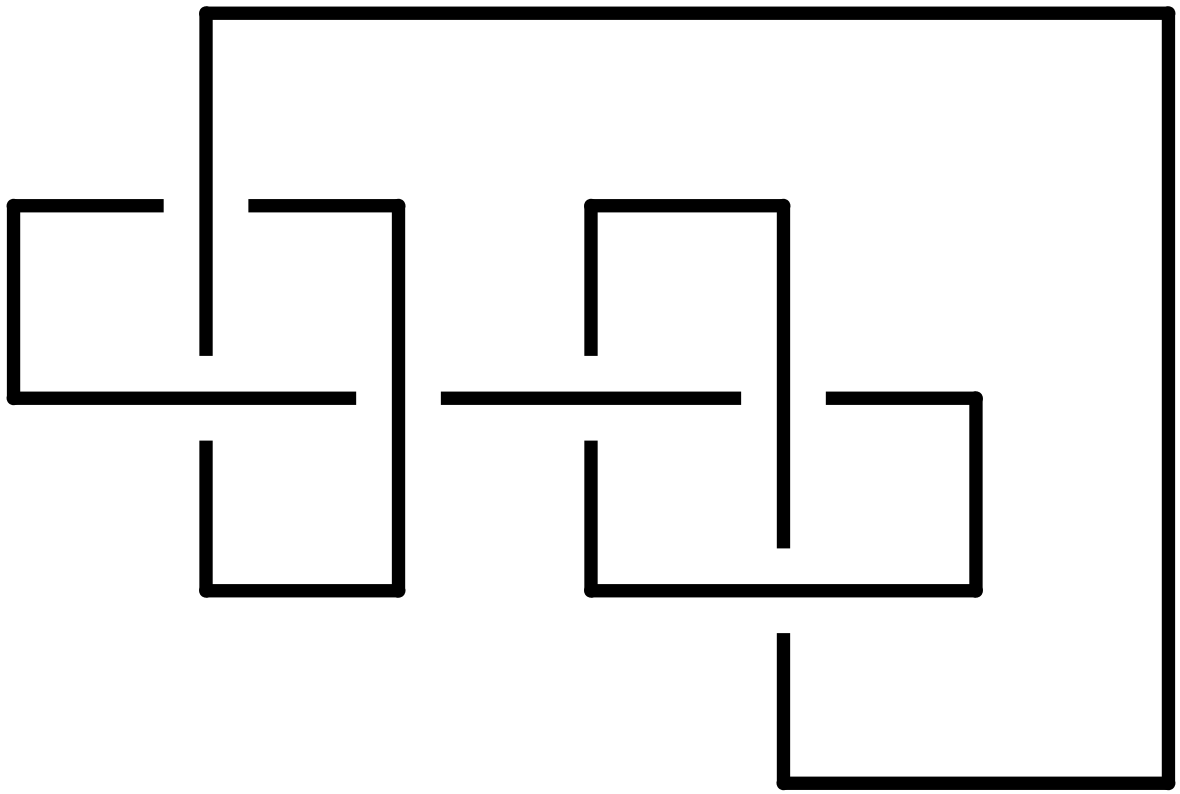


Gap = this strand passes UNDER the other one. Join the ends so the loop is closed.

Photos: c15_sA_v1.jpg, c15_sA_v2.jpg, c15_sA_v3.jpg, c15_sC_v1.jpg, c15_sC_v2.jpg, c15_sC_v3.jpg

c16: square (reef) knot

type 3_1 # 3_1* | material: rope 8 mm, any, about 8.0 mm | sessions: sB, sD

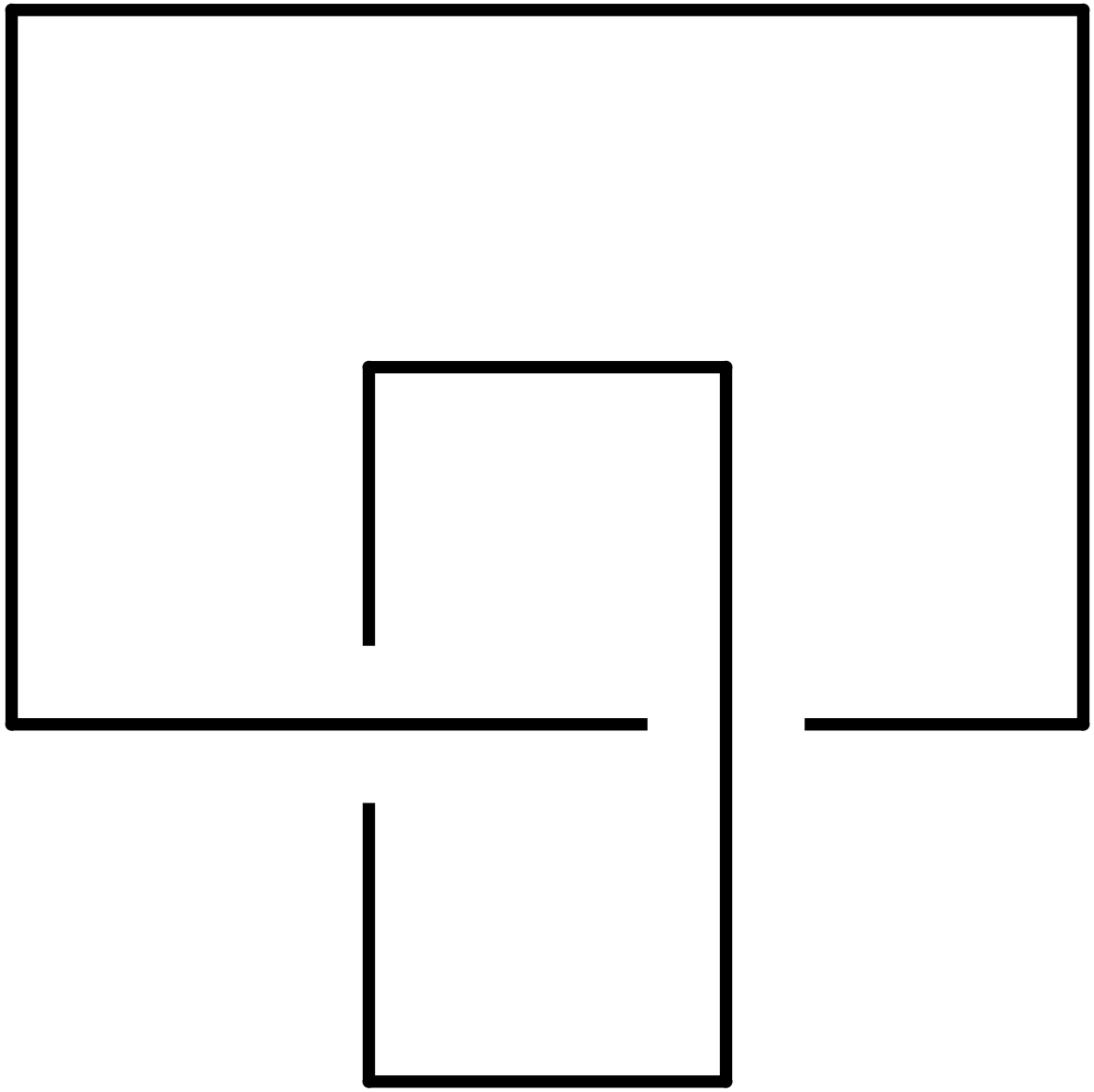


Gap = this strand passes UNDER the other one. Join the ends so the loop is closed.

Photos: c16_sB_v1.jpg, c16_sB_v2.jpg, c16_sB_v3.jpg, c16_sD_v1.jpg, c16_sD_v2.jpg, c16_sD_v3.jpg

c17: Hopf link (two linked cables)

type link[0_1,0_1|interlinked|lk=[[0, 1], [1, 0]]] | material: coax RG6 + patch cable (two cables), two colours, about 6.5 mm

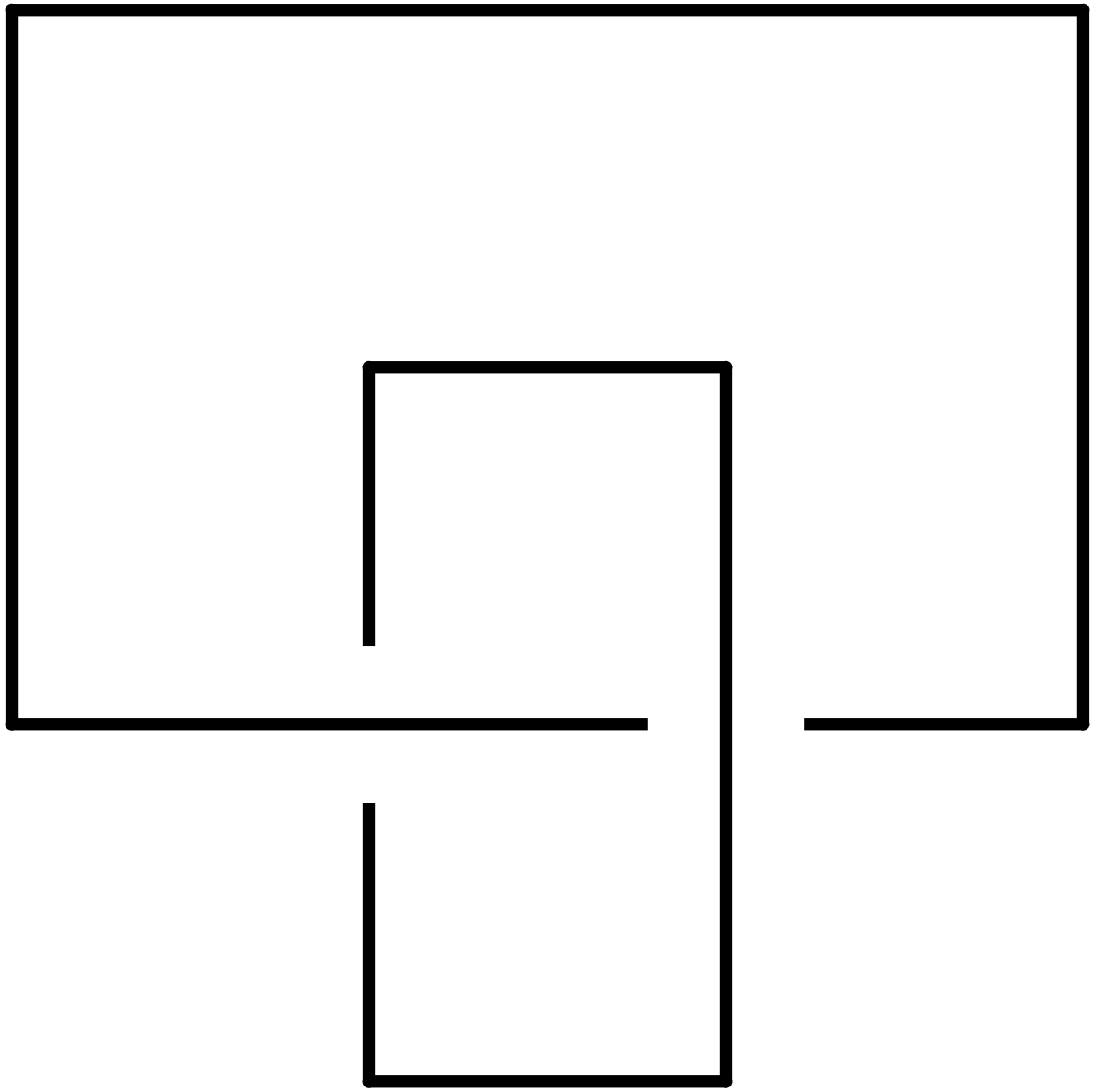


Gap = this strand passes UNDER the other one. Join the ends so the loop is closed.

Photos: c17_sA_v1.jpg, c17_sA_v2.jpg, c17_sA_v3.jpg, c17_sD_v1.jpg, c17_sD_v2.jpg, c17_sD_v3.jpg

c18: Hopf link (two linked cables)

type link[0_1,0_1|interlinked|lk=[[0, 1], [1, 0]]] | material: rope + thread (two cables), two colours, about 4.0 mm | session

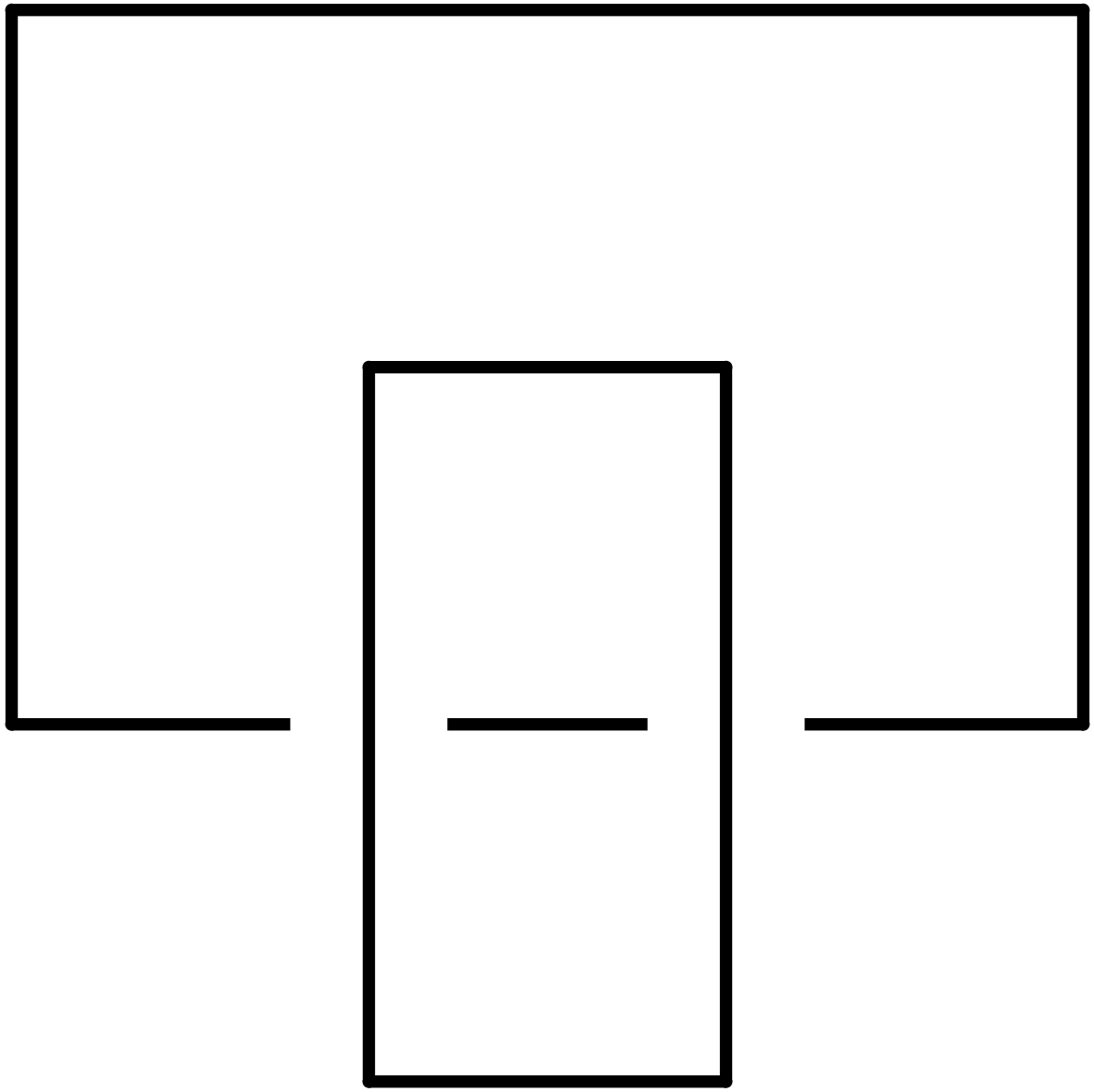


Gap = this strand passes UNDER the other one. Join the ends so the loop is closed.

Photos: c18_sB_v1.jpg, c18_sB_v2.jpg, c18_sB_v3.jpg, c18_sC_v1.jpg, c18_sC_v2.jpg, c18_sC_v3.jpg

c19: two separate cables crossing twice (unlinked)

type link[0_1,0_1|separable|lk=[[0, 0], [0, 0]]] | material: two coax cables crossing twice, not linked, any, about 6.9 mm |

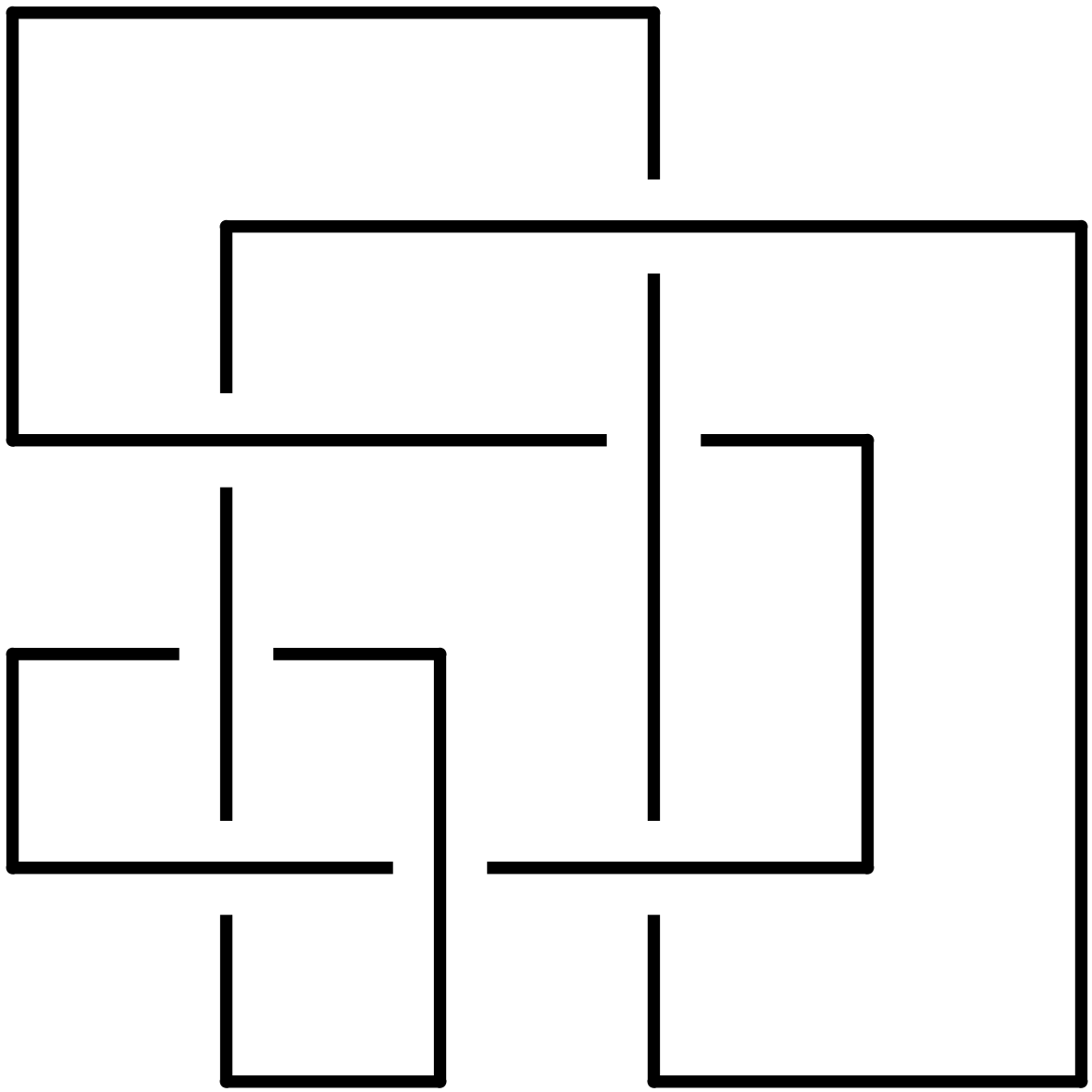


Gap = this strand passes UNDER the other one. Join the ends so the loop is closed.

Photos: c19_sA_v1.jpg, c19_sA_v2.jpg, c19_sA_v3.jpg, c19_sB_v1.jpg, c19_sB_v2.jpg, c19_sB_v3.jpg

c20: trefoil + figure-eight (connected sum)

type 3_1 # 4_1* | material: coax RG58, black, about 5.0 mm | sessions: sC, sD



Gap = this strand passes UNDER the other one. Join the ends so the loop is closed.

Photos: c20_sC_v1.jpg, c20_sC_v2.jpg, c20_sC_v3.jpg, c20_sD_v1.jpg, c20_sD_v2.jpg, c20_sD_v3.jpg